KAIZEN MATHS

Lesson starter

Straight Line Graphs and Gradients

Line Intersections

Name

S1

Find the intersection point of the lines:
y = x - 4
y = -3x + 4

S2

Find the equation of the line perpendicular to y = -3x - 2 passing through (3, 4)

S3

Find the intersection point of the lines:
y = -2x - 2
y = 3x - 5

S4

Find the equation of the line perpendicular to y = -3x + 5 passing through (-1, 0)

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ALG-FCA4-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Straight Line Graphs and Gradients

Line Intersections

Name _____

E1

Find the intersection point of the lines:
$y = -2x + 3$
$y = -3x - 4$

E2

Find the equation of the line perpendicular to $y = 2x - 5$ passing through $(1, 0)$

E3

Find the intersection point of the lines:
$y = -3x + 1$
$y = x + 3$

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Exit ticket

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Line Intersections

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Find the intersection point of the lines:
$y = -2x + 3$
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Find the equation of the line perpendicular to $y = 2x - 5$ passing through $(1, 0)$

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Find the intersection point of the lines:
$y = -3x + 1$
$y = x + 3$

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Straight Line Graphs and Gradients

Line Intersections

Name _____

E1

Find the intersection point of the lines:
$y = -2x + 3$
$y = -3x - 4$

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Find the equation of the line perpendicular to $y = 2x - 5$ passing through $(1, 0)$

E3

Find the intersection point of the lines:
$y = -3x + 1$
$y = x + 3$

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Teacher answers & working

Straight Line Graphs and Gradients · Line Intersections

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S1

Find the intersection point of the lines:

$y = x - 4$

$y = -3x + 4$

Answer:

Intersection point: (2, -2)

Working:

Set the equations equal to each other.

$x - 4 = -3x + 4$

$4x = 8$

$x = 2$

Substitute back into the first equation.

$y = -2$

Intersection point: (2, -2)

S2

Find the equation of the line perpendicular to y = -3x - 2 passing through (3, 4)

Answer:

$y = \frac{1}{3}x + 3$

Working:

The gradient of the given line is -3.

Perpendicular gradients multiply to -1, so the new gradient is $\frac{1}{3}$.

Use y = mx + c with point (3, 4).

$4 = \frac{1}{3}(3) + c$

$c = 3$

$y = \frac{1}{3}x + 3$

Teacher answers & working

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S3

Find the intersection point of the lines:

y = -2x - 2

y = 3x - 5

Answer:

Intersection point: $(\frac{3}{5}, \frac{-16}{5})$

Working:

Set the equations equal to each other.

$-2x - 2 = 3x - 5$

$-5x = -3$

$x = \frac{3}{5}$

Substitute back into the first equation.

$y = \frac{-16}{5}$

Intersection point: $(\frac{3}{5}, \frac{-16}{5})$

S4

Find the equation of the line perpendicular to y = -3x + 5 passing through (-1, 0)

Answer:

$y = \frac{1}{3}x + \frac{1}{3}$

Working:

The gradient of the given line is -3.

Perpendicular gradients multiply to -1, so the new gradient is $\frac{1}{3}$.

Use y = mx + c with point (-1, 0).

$0 = \frac{1}{3}(-1) + c$

$c = \frac{1}{3}$

$y = \frac{1}{3}x + \frac{1}{3}$

Teacher answers & working

Straight Line Graphs and Gradients · Line Intersections

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E1

Find the intersection point of the lines:

y = -2x + 3

y = -3x - 4

Answer:

Intersection point: (-7, 17)

Working:

Set the equations equal to each other.

$-2x + 3 = -3x - 4$

$x = -7$

$x = -7$

Substitute back into the first equation.

$y = 17$

Intersection point: (-7, 17)

E2

Find the equation of the line perpendicular to y = 2x - 5 passing through (1, 0)

Answer:

$y = -\frac{1}{2}x + \frac{1}{2}$

Working:

The gradient of the given line is 2.

Perpendicular gradients multiply to -1, so the new gradient is $\frac{-1}{2}$.

Use y = mx + c with point (1, 0).

$0 = \frac{-1}{2}(1) + c$

$c = \frac{1}{2}$

$y = -\frac{1}{2}x + \frac{1}{2}$

Teacher answers & working

Straight Line Graphs and Gradients · Line Intersections

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E3

Find the intersection point of the lines:

y = -3x + 1

y = x + 3

Answer:

Intersection point: $(\frac{-1}{2}, \frac{5}{2})$

Working:

Set the equations equal to each other.

$-3x + 1 = x + 3$

$-4x = 2$

$x = \frac{-1}{2}$

Substitute back into the first equation.

$y = \frac{5}{2}$

Intersection point: $(\frac{-1}{2}, \frac{5}{2})$